LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data
Student: Ranjitha R
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Objective	**Dataset**	**Model**
Predict Outcome: 0 = No diabetes, 1 = Diabetes.	Pima Indians Diabetes Database from Kaggle; file: diabetes.csv.	Median imputation + StandardScaler + balanced Logistic Regression.

Dataset and Workflow

Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Workflow: load Kaggle CSV → inspect data → replace invalid zeros → train-test split → median imputation → scaling → train Logistic Regression → evaluate → interpret coefficients.

Evaluation Metrics

Metric	Result	Meaning
Accuracy	0.734	Overall correct predictions
Precision	0.603	Reliability of positive predictions
Recall	0.704	Actual positive cases detected
F1-score	0.650	Balance of precision and recall
ROC-AUC	0.813	Overall class-separation ability

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

Conclusion

In this project, I built a Logistic Regression model to predict diabetes using the Pima Indians dataset, with median imputation, scaling, and balanced class weights. The model reached a ROC-AUC of 0.813 and a recall of 0.704, meaning it correctly detected about 70% of actual diabetic cases. Glucose and BMI were the strongest predictors of diabetes. Recall matters most in a screening task because a missed case (false negative) is more costly than a false alarm, though the precision of 0.603 shows that some non-diabetic patients were flagged (25 false positives). Since the dataset is small and comes from one specific population, this model is useful for learning classification concepts but not for real medical diagnosis.